

Aryan Sharma

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SUMMARY

Software Engineer with hands-on **C++** and **Linux** experience in safety-critical industrial systems, delivering production-grade middleware components and automated testing frameworks. Proficient in **object-oriented design, multithreading concepts, and Linux system programming**, with a track record of independently owning and executing core software modules.

SKILLS

Languages	C, C++/C++11, Python, Bash
Systems & Middleware	Linux system programming, IPC mechanisms (sockets, pipes, shared memory), REST/HTTP APIs (Oat++ framework), modular middleware architecture, multithreading concepts
OOP & Design	Object-oriented design, design patterns (Factory, Strategy, Observer), SOLID principles, modular component architecture
Dev & Test Tools	GDB, Git, GitHub CI/CD, JIRA, CMake, Valgrind (basic), unit testing, TDD practices, Python (automation & scripting)
Embedded & Hardware	STM32 microcontrollers, SPI/I2C/UART protocols, bare-metal C, hardware debugging, oscilloscope usage

WORK EXPERIENCE

Graduate Engineer Trainee — Industrial Software | *Alstom Transport India Ltd., Bangalore* Aug 2025 – Apr 2026

- Independently designed and owned a **modular C++ middleware validation engine** (Oat++ REST backend) for safety-critical railway control data — structured around **OOP design patterns** (Factory, Strategy) with configurable rule sets, enabling reuse across multiple data format modules.
- Implemented **multi-stage data processing pipelines** in C++ handling structured message parsing, rule evaluation, and error reporting; applied **C++11 features** (lambdas, smart pointers, move semantics) throughout for robust resource management.
- Built **Python regression automation scripts** integrated into a **Git-based CI/CD pipeline**, cutting manual QA effort by **~40%** — experience directly applicable to **TDD and component test** workflows.
- Collaborated with cross-functional global teams across architecture, integration, and QA; tracked and resolved defects using **JIRA**, contributing to requirements analysis and design review discussions.
- Worked in a **Linux development environment** daily — build systems, shell scripting, process management, and **GDB-based debugging** of C++ runtime issues.

PROJECTS

Middleware-Style C++ Rule Engine — Configurable Data Validation Platform *C++11, Oat++, REST, Python*

- Architected a **plugin-style middleware component** in C++ with a clean separation between the rule-loader, evaluation engine, and reporting layer — each independently testable and swappable.
- Exposed functionality as a **RESTful HTTP API** (Oat++ framework on Linux), enabling downstream integration with other platform services — analogous to **IPC/RPC inter-service communication** in IVI middleware stacks.
- Applied **multithreading** for concurrent rule evaluation across independent data streams; used **mutexes and condition variables** (C++11 <thread>, <mutex>) to manage shared state safely.

Transformer Predictive Maintenance — ML + Signal Processing Pipeline *Python, XGBoost, SHAP, Streamlit*

- Designed an end-to-end software pipeline from raw sensor ingestion to fault classification, achieving **94% fault detection accuracy** on class-imbalanced real-world data using XGBoost and Random Forest.
- Structured the pipeline as **decoupled software modules** (ingest, feature extraction, model inference, reporting) — reinforcing **component ownership and modular software architecture** practices.
- Deployed as a **Streamlit web application** with live inference, demonstrating full software development lifecycle from prototype to production delivery.

STM32 Sensor Data Logger — Embedded Linux Integration *C, STM32, SPI/I2C/UART, Python*

- Built a **bare-metal C firmware** for STM32 sampling IMU + environmental sensors over **SPI/I2C**, streaming timestamped data via UART to a **Linux host** for real-time processing.
- Wrote a **Linux-side Python daemon** to receive, parse, and log serial data — involving serial port IPC, file I/O, and signal handling in a Linux environment.
- Debugged hardware-software integration issues using **oscilloscope and logic analyser**, resolving SPI clock polarity and I2C address conflicts — hands-on hardware debugging experience relevant to **IVI platform bring-up**.

EDUCATION

2021 – 2025 B.Tech in Electrical Engineering — **NIT Hamirpur** (CGPA: 7.91/10)
Relevant Coursework: **Data Structures & Algorithms, Operating Systems, Digital Electronics**, Embedded Systems, Digital Signal Processing, Computer Networks

ACHIEVEMENTS & COURSES

Software Achievement Independently designed, owned, and delivered a modular C++ middleware validation engine in a production industrial software environment — from architecture to CI/CD integration.

ML Achievement Achieved **94% fault detection accuracy** on class-imbalanced real-world data; end-to-end pipeline from raw sensor signals to deployed web inference app.

Courses Complete Data Science, ML, DL & NLP Bootcamp — *Krish Naik, KRISHAI Technologies (Udemy)*
Complete Agentic AI Bootcamp with LangGraph & LangChain — *Krish Naik, KRISHAI Technologies (Udemy)*
Linux System Programming & C++ Concurrency — *(actively pursuing)*